CSCI 5526

Unsupervised Problems in Healthcare

Announcements

- MP1 due 9/25 at 11.59pm.
- MP2 will be released Monday 9/28.
- Finalize project groups by 10/16.

Student Questions

Are deep learning models being over utilized in situations where a classic ML model could have in a simpler way solved the same problem?

-- Yes, there is obviously a bias towards using DL for all tasks because they are more modular.

-- But DL certainly has advantages when the data size is large.

When different healthcare modalities provide conflicting information—for example, imaging suggesting one diagnosis while clinical notes or lab results suggest another—how should a multimodal model learn which modality to trust?

-- They weigh each modality based on learned clinical context, historical reliability, and task-specific relevance

-- some systems calculate predictive uncertainty estimates when there is conflicting information

Student Questions

Why do feature selection before training a model if we could just use regularization like l1,l2?

-- Regularization-based feature selection is performed when manual feature selection is not obvious.

Is there a good method or process of analysis that can help decide what model(s) to use for a specific problem? Especially when there are multiple viable options given the data type, do ML scientists usually experiment with all of the models before moving on to training with the best one?

-- There is not a simple answer. There are some factors to consider, but even then, the answer may not be straight forward.

How to choose a model?

- Assumptions
  - Logistic/linear regression, linear SVM: linear separability
- Feature types:
  - Tree-based models: suited for categorical features
- Interpretability
  - Logistic/linear regression, linear SVM: coefficients
  - Tree ensembles: variable importance
  - Caveat: co-linear features

Student Questions

If a healthcare model performs well on the training data but poorly on unseen patients how can we determine whether the main problem is high variance from overfitting or high bias from incorrect assumptions from the patient population?

-- See whether the test set performance improves when you reduce model complexity

-- Also compare internal validation set performance

Since the future of machine learning with complex data types (e.g. audio, text, images) seems to be the use of pre-trained foundation models with a simple head to fine-tune it to the task at hand, can we spend more of the course on this topic?

-- Yes we will spend the latter half of this class on this topic.

Student Questions

I would like to understand how changing Bias can affect performance of ML model if possible.

-- Improving bias means you are using a more capable model to capture the data characteristics.

If a model performs well during validation but not as well when deployed at a different hospital, how do we determine if the problem is dataset shift, differences in clinical practice, or bias in the original training data?

-- This certainly requires an audit. May need a closer inspection of the data and consultation with clinical experts.

How do we know that imputation is correcting the missing information or is it introducing new biases or shortcuts for the model?

-- One useful trick is to include a feature Is_Feature_X_Missing (0 or 1) after imputing and see if the model heavily relies on that feature.

Outline

- Goals of unsupervised learning
- Relevance to healthcare
- Examples in healthcare
- Principal Component Analysis
- Independent Component Analysis

Unsupervised Learning

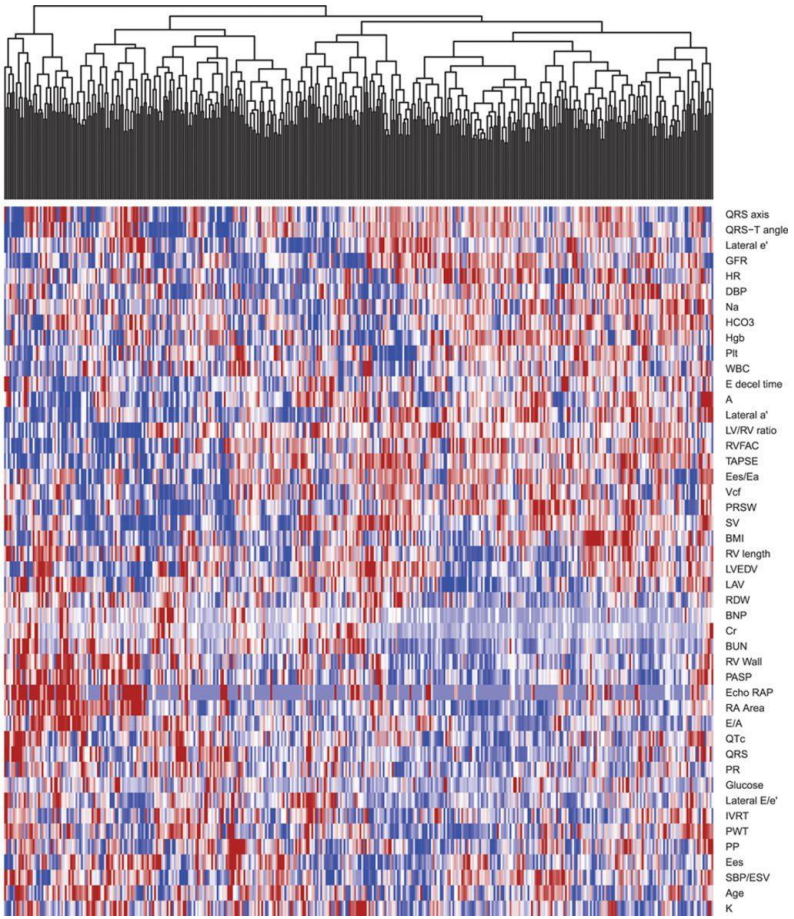
- Learning structure in data without expert labels
- Main goals:
  - Clustering and density estimation
    - Mixture models, DBSCAN
    - K-means, hierarchical clustering
  - Structure discovery (reduction or transformation)
    - PCA, ICA, t-SNE, tensor decomposition
  - Representation learning/feature encoders
    - Autoencoders, self-supervised learning
  - Generative models
    - GANs, diffusion models

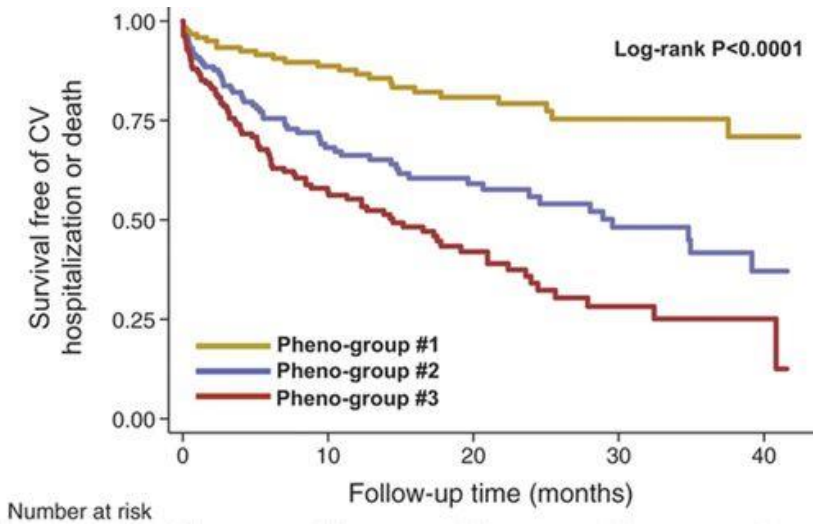
Why is it important in healthcare?

- Labels are costly
  - Experts must label the examples
  - Time-consuming
- Labeling inconsistencies
  - Disagreement between experts
  - Erroneous labels (due to burnout)
- Discovery
  - Sometimes we do not know the labels

Examples in Healthcare

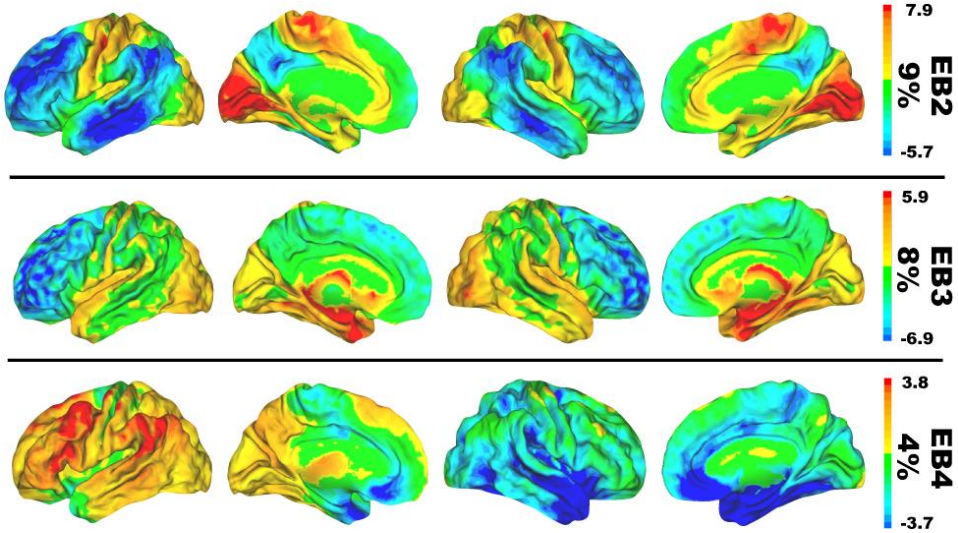
Subtype Discovery in Heart Failure

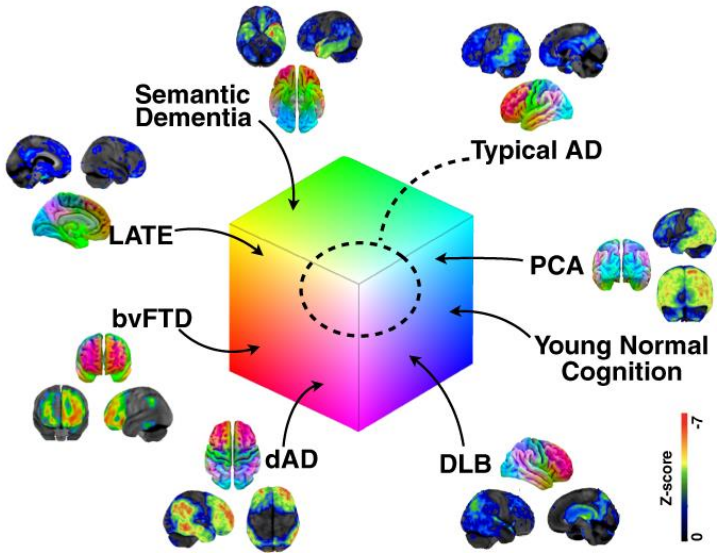




Source: https://doi.org/10.1161/CIRCULATIONAHA.114.010637

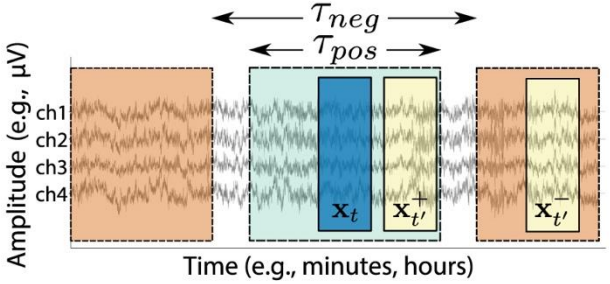
Profiling Cognitive Diseases using PCA of Brain PET Images

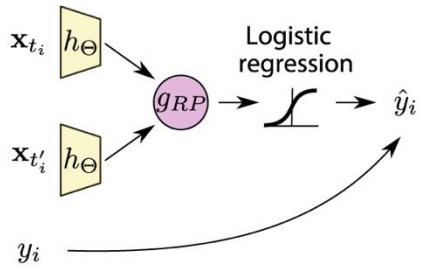


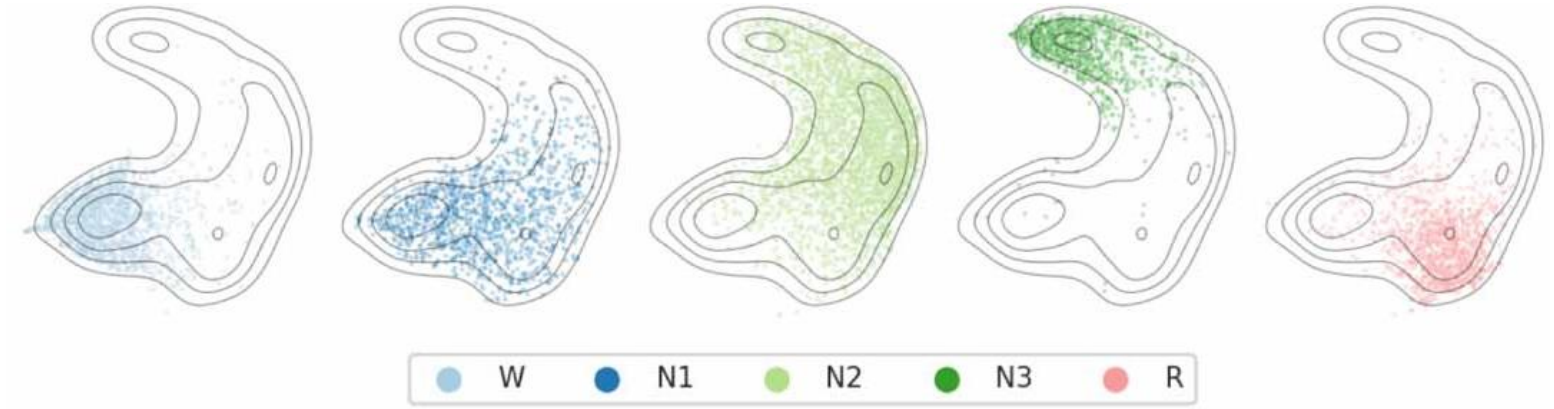


Source: https://doi.org/10.1038/s41467-022-29047-4

Self-supervised Learning of EEG Data







Source: https://iopscience.iop.org/article/10.1088/1741-2552/abca18/

MED2VEC: Embedding Clinical Concepts using Self-supervised Learning

Medical codes with the strongest value in six different coordinates of the 200-dimensional code embedding space.

Coordinate 112	Coordinate 152	Coordinate 141
Kidney replaced by transplant (V42.0)	X-ray, knee (P)	Cystic fibrosis (277.02)
Hb-SS disease without crisis (282.61)	X-ray, thoracolumbar (P)	Intracranial injury (854.00)
Heart replaced by transplant (V42.1)	Accidents in public building (E849.6)	Persistent mental disorders (294.9)
RBC antibody screening (P)	Activities involving gymnastics (E005.2)	Subdural hemorrhage (432.1)
Complications of transplanted	Struck by objects/persons in sports (E917.0)	Neurofibromatosis (237.71)
bone marrow (996.85)	Encounter for removal of sutures (V58.32)	Other conditions of brain (348.89)
Sickle-cell disease (282.60)	Struck by object in sports (E917.5)	Conductive hearing loss (389.05)
Liver replaced by transplant (V42.7)	Unspecified fracture of ankle (824.8)	Unspecified causes of encephalitis,
Hb-SS disease with crisis (282.62)	Accidents occurring in place for	myelitis, encephalomyelitis (323.9)
Prograf PO (R)	recreation and sport (E849.4)	Sensorineural hearing loss (389.15)
Complications of transplanted heart (996.83)	Activities involving basketball (E007.6)	Intracerebral hemorrhage (431)
Coordinate 184	**Coordinate 190**	**Coordinate 199**
	Down's syndrome (758.0)	
Pain in joint, shoulder region (719.41)	Congenital anomalies (759.89)	Infantile cerebral palsy (343.9)
Pain in joint, lower leg (719.46)	Tuberous sclerosis (759.5)	Congenital quadriplegia (343.2)
Pain in joint, ankle and foot (719.47)	Anomalies of larynx, trachea,	Congenital diplegia (343.0)
Pain in joint, multiple sites (719.49)	and bronchus (748.3)	Quadriplegia (344.00)
Generalized convulsive epilepsy (345.10)	Autosomal deletions (758.39)	Congenital hemiplegia (343.1)
Pain in joint, upper arm (719.42)	Conditions due to anomaly of unspecified	Baclofen 10mg tablet (R)
Cerebral artery occlusion (434.91)	chromosome (758.9)	Wheelchair management (P)
MRI, brain (780.59)	Acquired hypothyroidism (244.9)	Tracheostomy status (V44.0)
Other joint derangement (718.81)	Conditions due to chromosome anomalies (758.89)	Paraplegia (344.1)
Fecal occult blood (790.6)	Anomalies of spleen (759.0)	Baclofen 5mg/ml liquid (R)
	Conditions due to autosomal anomalies (758.5)	

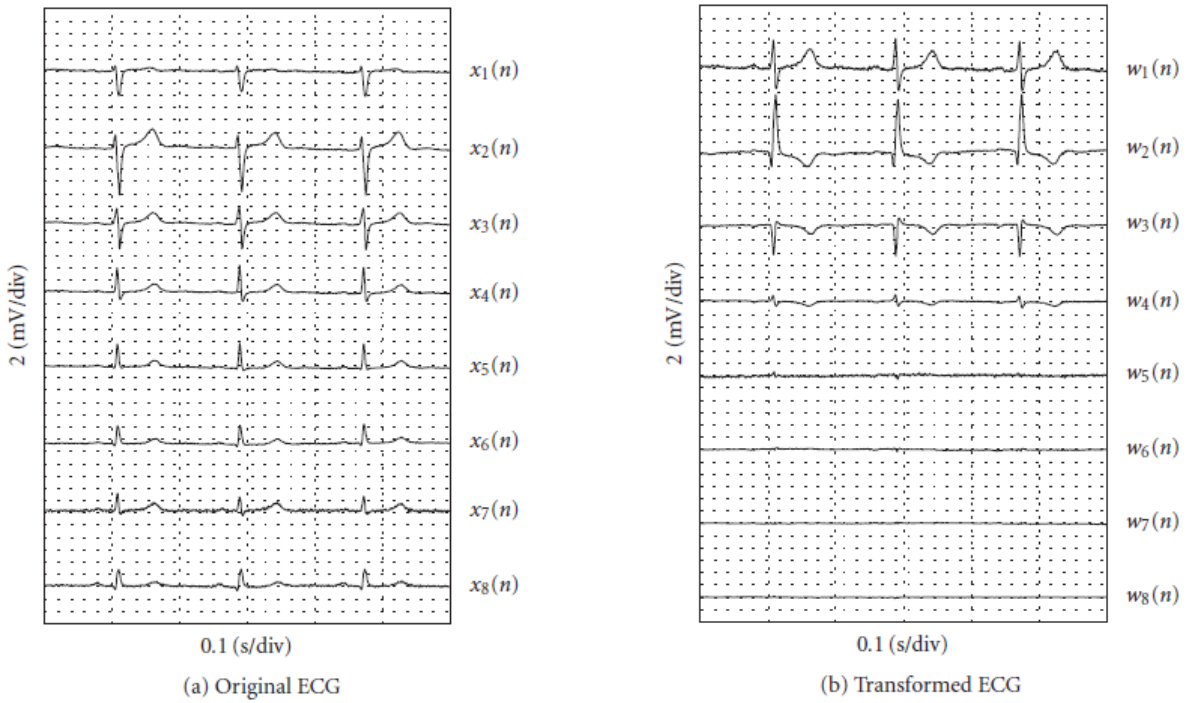
Source: https://www.kdd.org/kdd2016/papers/files/rpp0303-choiA.pdf

Unsupervised Structure Discovery: Principal Component Analysis (PCA)

Why PCA?

- Find interesting lower dimensional structure in data.
  - Extract few patterns that explain most of the data.

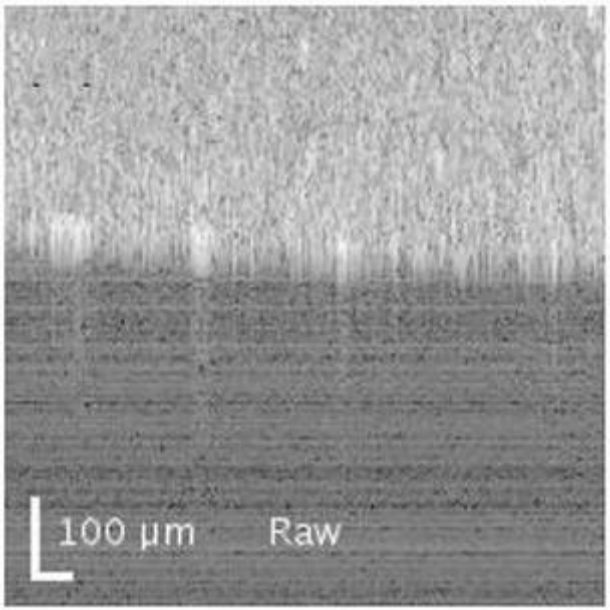
Finding "interesting" structure in data

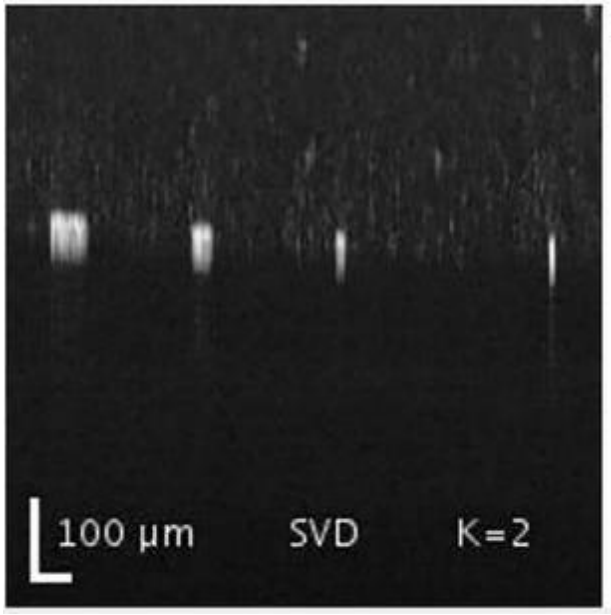


Example: Identifying representative patterns in multi-lead ECG signals

Castells, Francisco, et al. "Principal component analysis in ECG signal processing." EURASIP Journal on Advances in Signal Processing 2007 (2007): 1-21.

Finding "interesting" structure in data





Example: Separating static and blood flow signals in optical coherence tomography (OCT) data

Le, Nhan, et al. "Robust principal component analysis in optical micro-angiography." Quantitative imaging in medicine and surgery 7.6 (2017): 654.

Basics: Data Representation

- Suppose we take multiple measurements (n) of a phenomenon using multiple sensors (m)
  - E.g., measuring heart activity using multiple ECG leads

Measurements

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}$$ Sensors

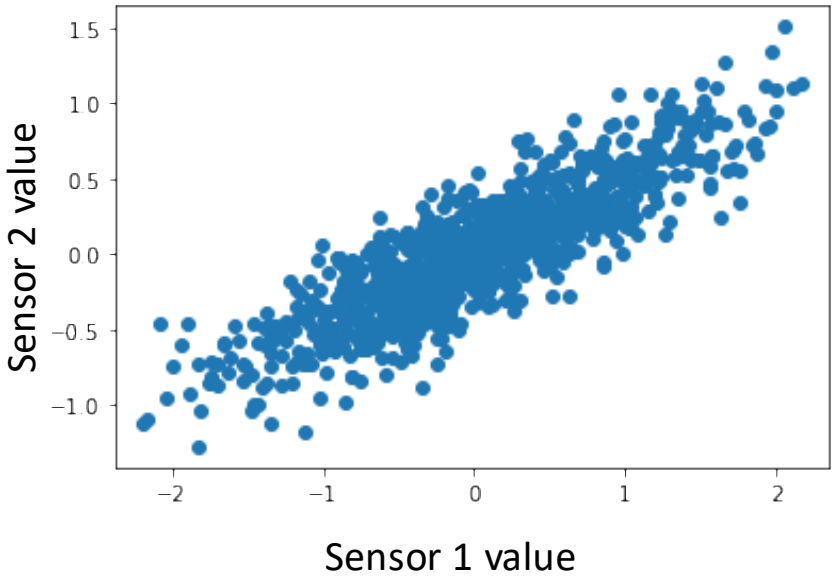
- We assume that the data is centered
  - The mean of each sensor measurement is 0, with some variance.

Goals of PCA

- Identify the sources of variance, such that
  - Signal to noise ratio is maximized
  - Redundancy between the sources is minimized

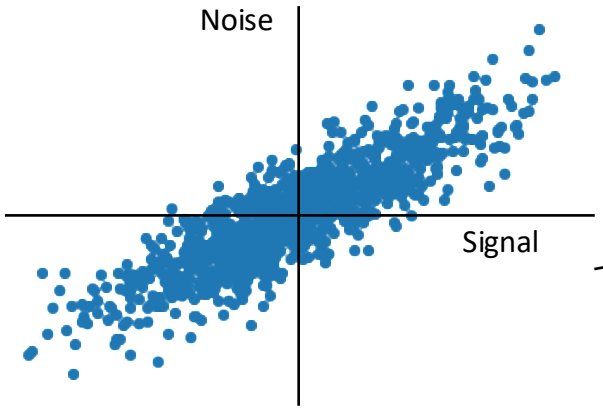
Example

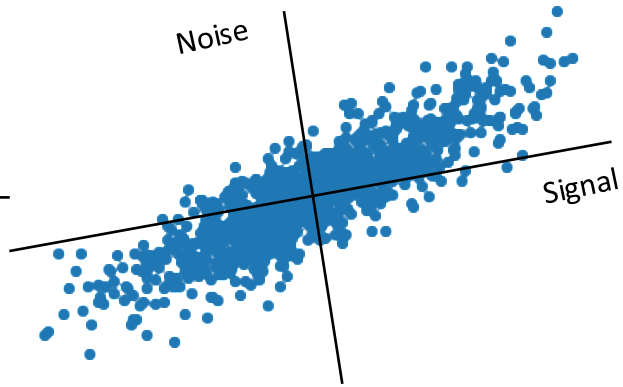
$$X = \begin{bmatrix} -0.38 & -0.80 & \cdots & 0.64 & 0.30 \\ -0.53 & -0.04 & \ldots & 0.20 & 0.21 \end{bmatrix}$$

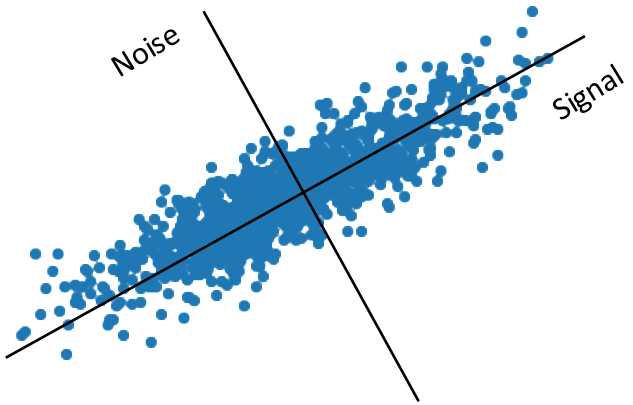


Defining "interesting" structure

- Directions with largest variance contain 'signals' of interest
- Directions with small variance represent 'noise' (and is orthogonal to 'signal')
- Maximize signal to noise ratio (SNR) $SNR = \frac{\sigma^2_{signal}}{\sigma^2_{noise}}.$

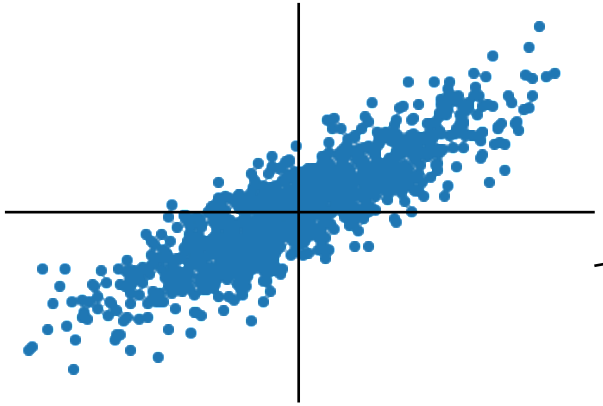




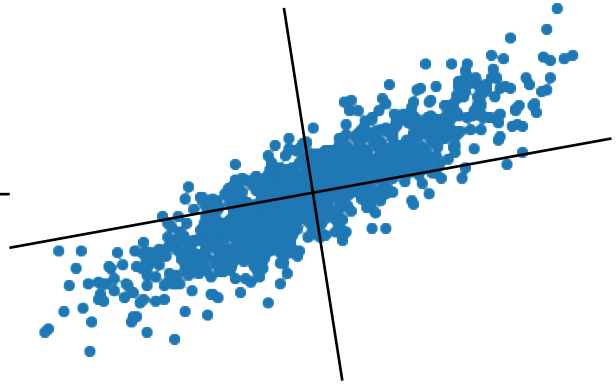


Redundancy

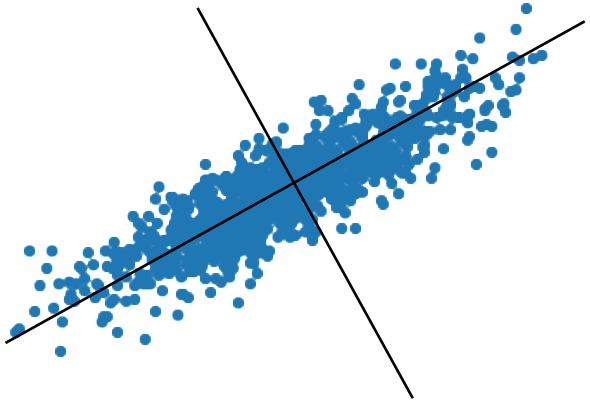
- Which of these bases carry redundant information?
  - Can we predict the one coordinate value from the other (are the two axes correlated)?
  - Correlation $\rho = \frac{1}{n}\sum_{i=1}^{n} x_{1i}.x_{2i} = \frac{1}{n}\mathrm{x}_1\mathrm{x}_2^{\mathrm{T}}$



$\rho > 0$



$\rho > 0$



$\rho = 0$

PCA Steps

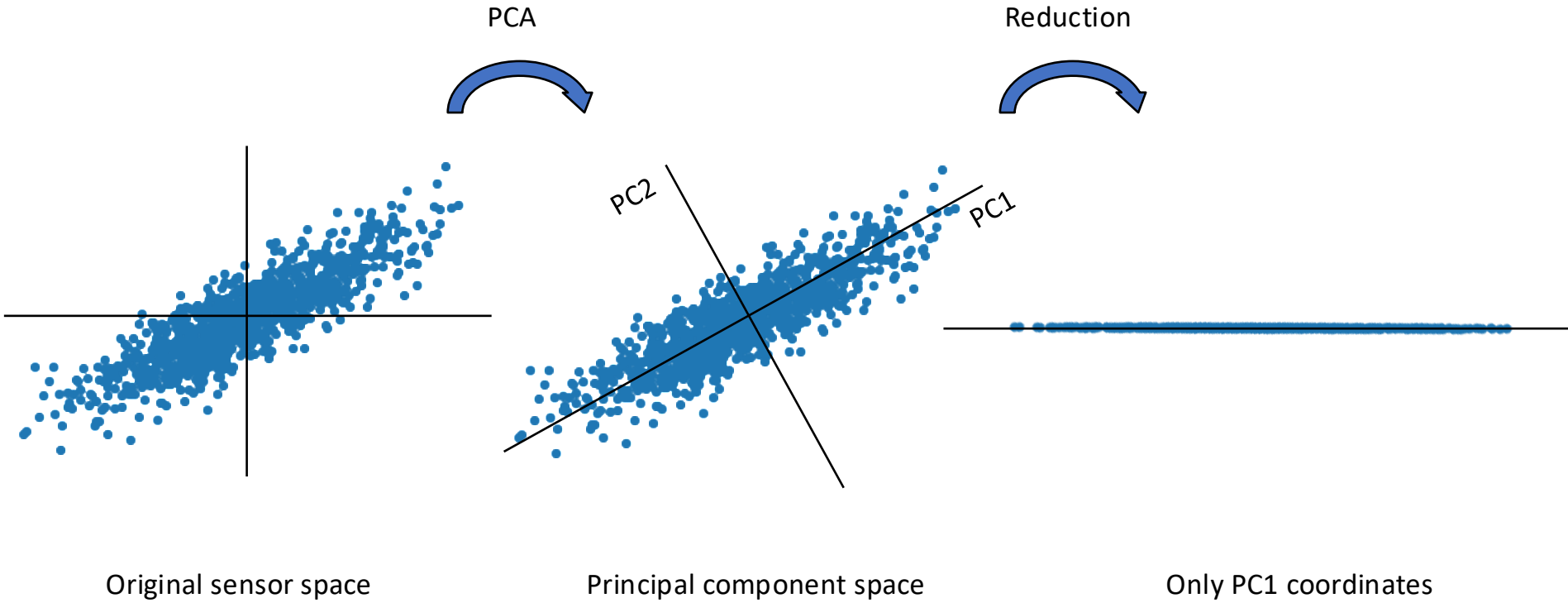
- Given centered data X
  - Rows are sensors and columns are measurements
  - Each column contains the coordinates of each measurement with respect to the original basis vectors (as defined by the sensors)
- Compute the covariance matrix (C_X) of X
- Perform eigenvalue decomposition of C_X
  - $C_X = E^T D E$ (E contains the eigenvectors of C_X in its rows)
- Define $P = E$
- Obtain new coordinates $Y = PX$
  - Y are the coordinates of the measurements with respect to the new basis

PCA Interpretation

- The eigenvectors of C_X (or rows of P) are called principal components (PCs).

- Each eigenvalue of C_X represents the variance explained by the corresponding eigenvector (or PC).

Retrieving interesting signal using PCA



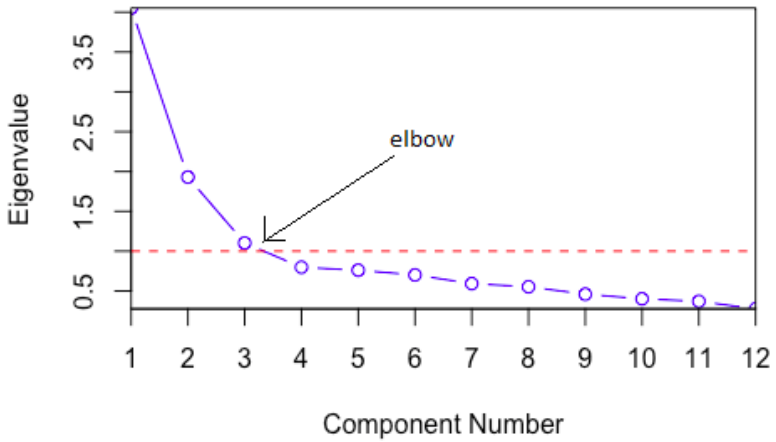
Original sensor space

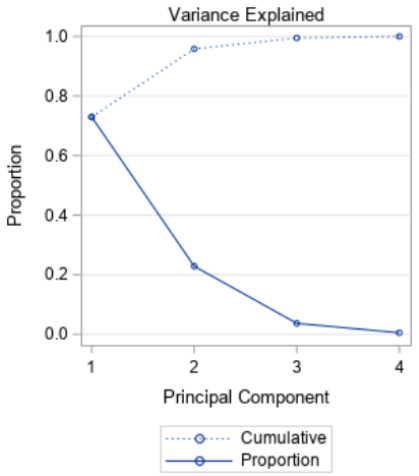
Principal component space

Only PC1 coordinates

Which PCs to keep when dim > 2?

- Scree plot – how many PCs to retain?
  - Eigenvalue = amount of variation the PC captures from the data
  - Popular heuristics
    - Elbow rule - look for the 'elbow' in the plot
    - Kaiser rule - pick PCs with eigenvalues of at least 1
    - Proportion of variance plot - the selected PCs should be able to describe at least 80% of the variance.





Dimensionality reduction using PCA

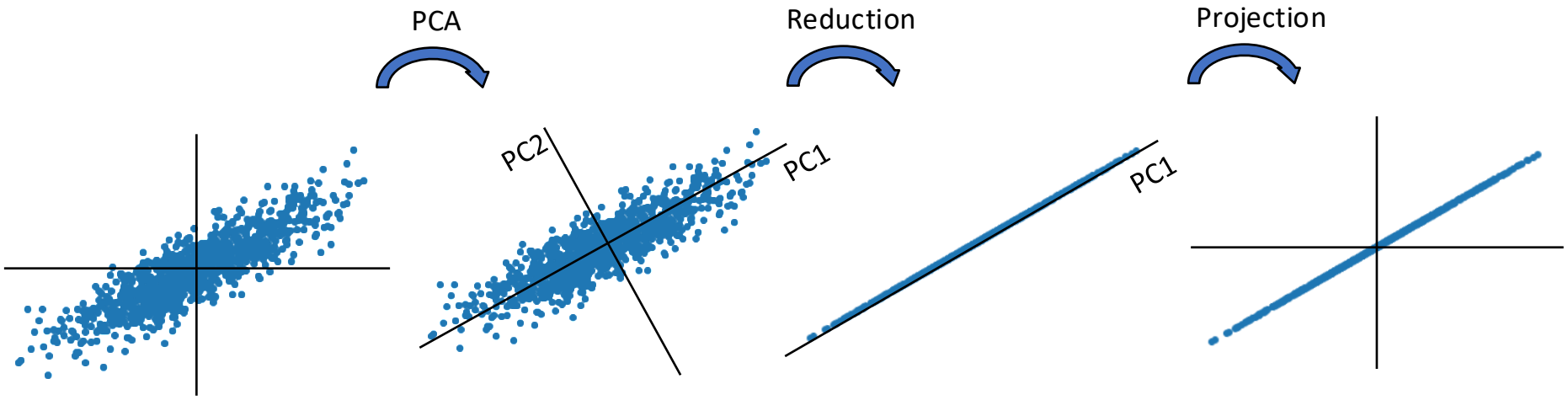
- Construct $\mathbf{P_k}$: re-arrange $\mathbf{k}$ rows of $\mathbf{P}$
  - $\mathbf{k}$ eigenvectors corresponding to the $\mathbf{k}$-largest eigenvalues of $\mathbf{C_X}$ (ordered)

- k-dimensional representation of $\mathbf{X}$ is given by: $\mathbf{P_k X}$

- In Scikit Learn:

```
pca = PCA(n_components=0.95, svd_solver='full')
pca.fit(X)
X_projected = pca.transform(X)
```

Keeping the interesting data in the original sensor space



Original sensor space

Principal component space

Reduced principal component space

Data projected onto PC1 in the original sensor space

How to keep data in the original space after dimensionality reduction?

- Perform projection:

$$X_{reduced} = P_k^T X_{proj}$$

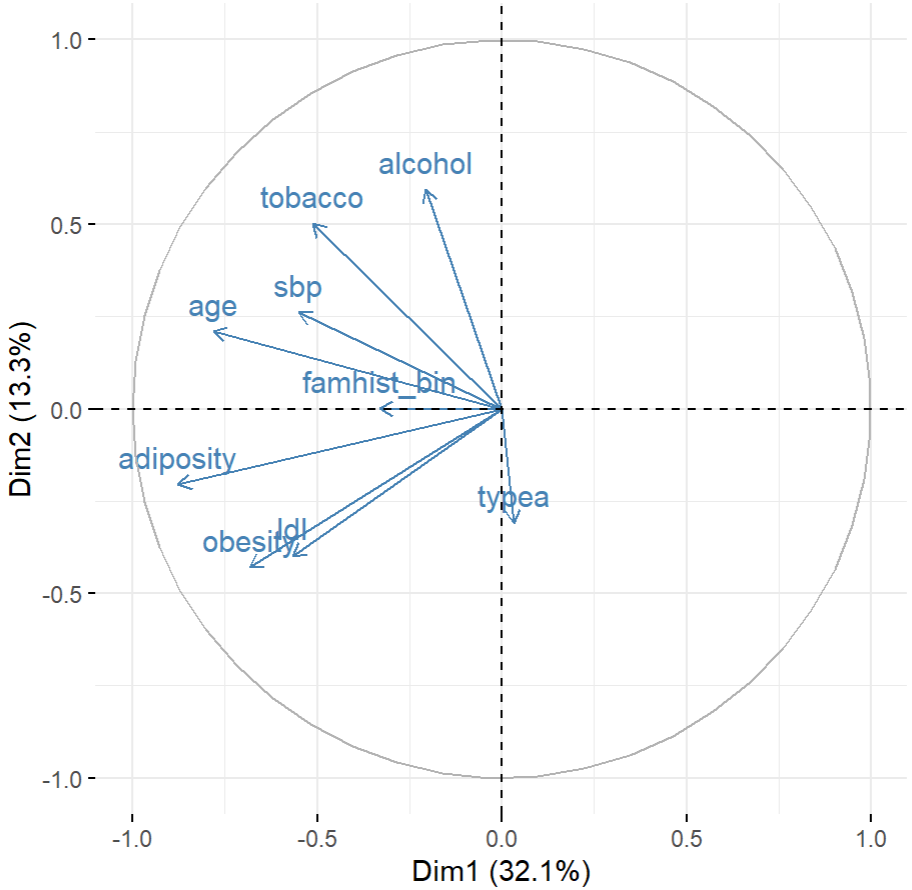
- In Scikit Learn:

```
X_orig = pca.inverse_transform(X_projected)
```

Interpreting and visualizing PCs

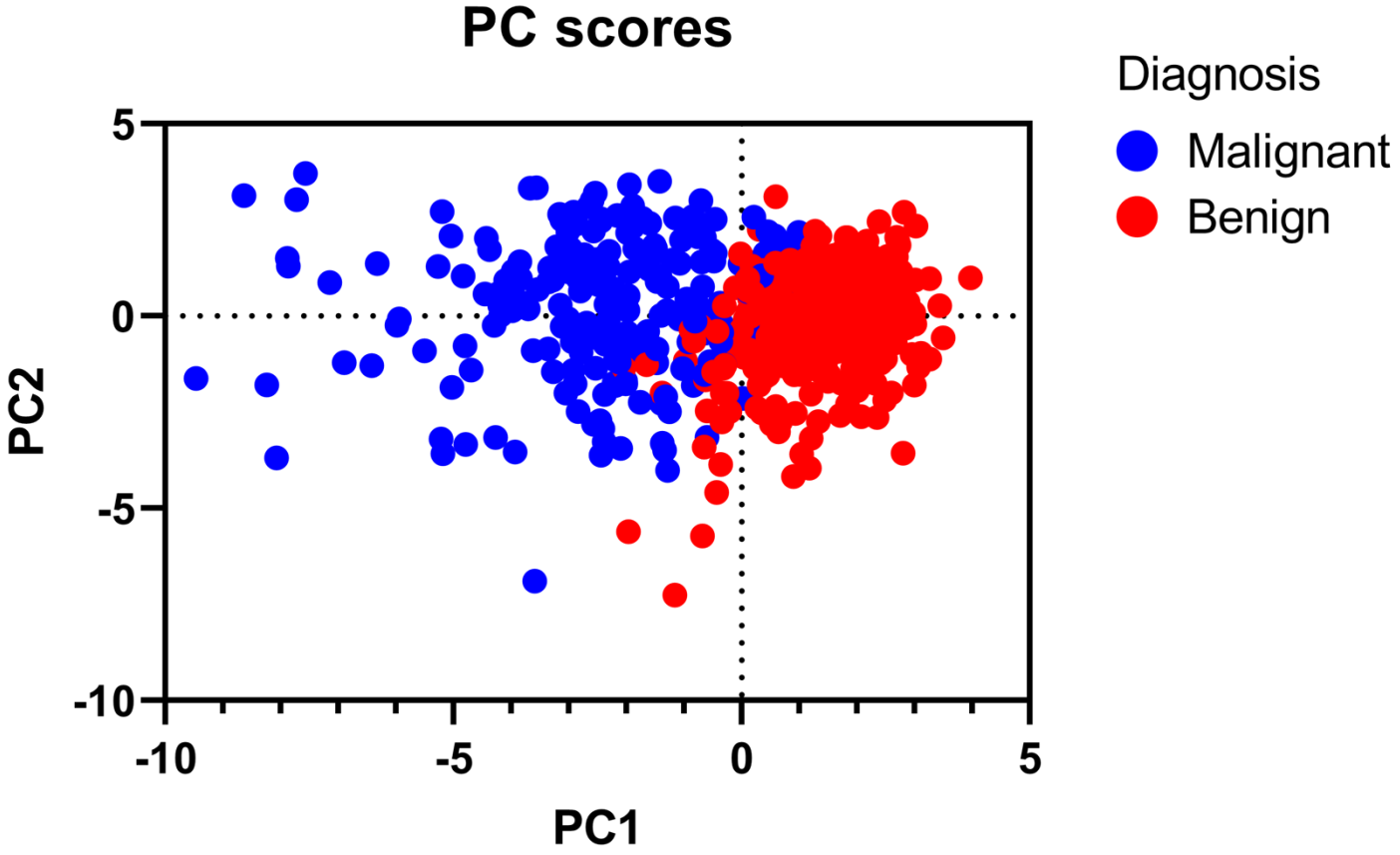
- Loadings = coefficients in eigenvectors of C_X
- Help understand how original variables contribute to each of the PCs
  - Positive loadings indicate that a variable and a principal component are positively correlated: an increase in one results in an increase in the other.
  - Negative loadings indicate a negative correlation.
  - Large (either positive or negative) loadings indicate that a variable has a strong effect on that principal component.
  - Angles between vectors indicates correlation between them
- Loadings plot – plot of coefficients of original variables in each PC

Loadings for Heart Disease Data



Source: https://rpubs.com/krzysztofkalisiak/872031

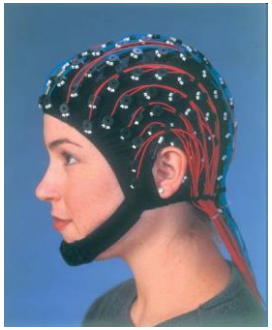
Visualizing the Reduced Data

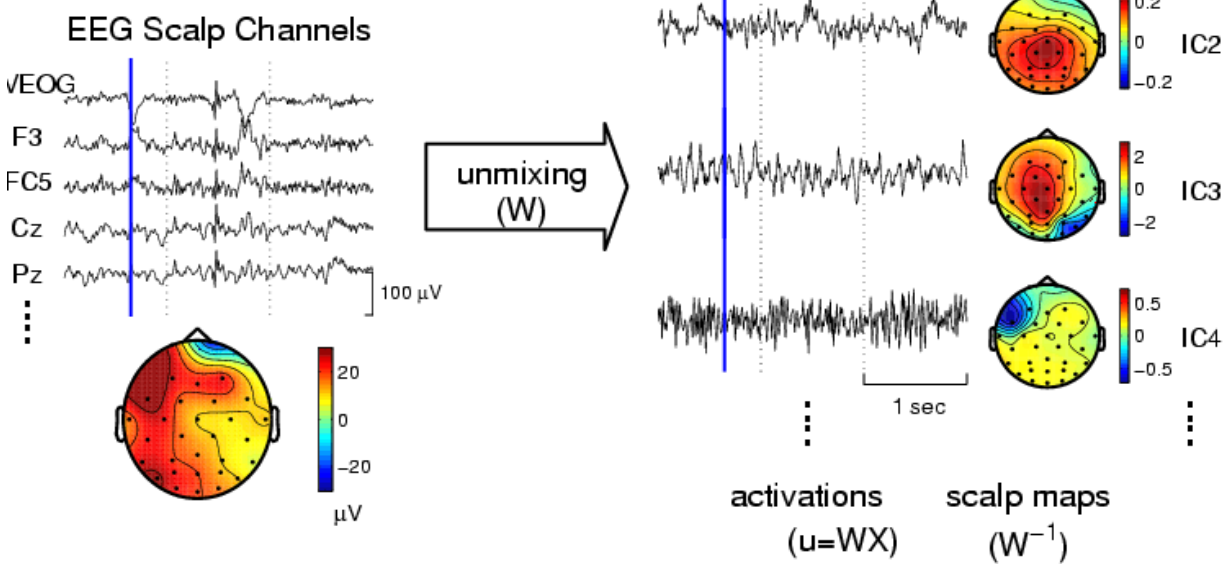


ICA: Cocktail-party Problem

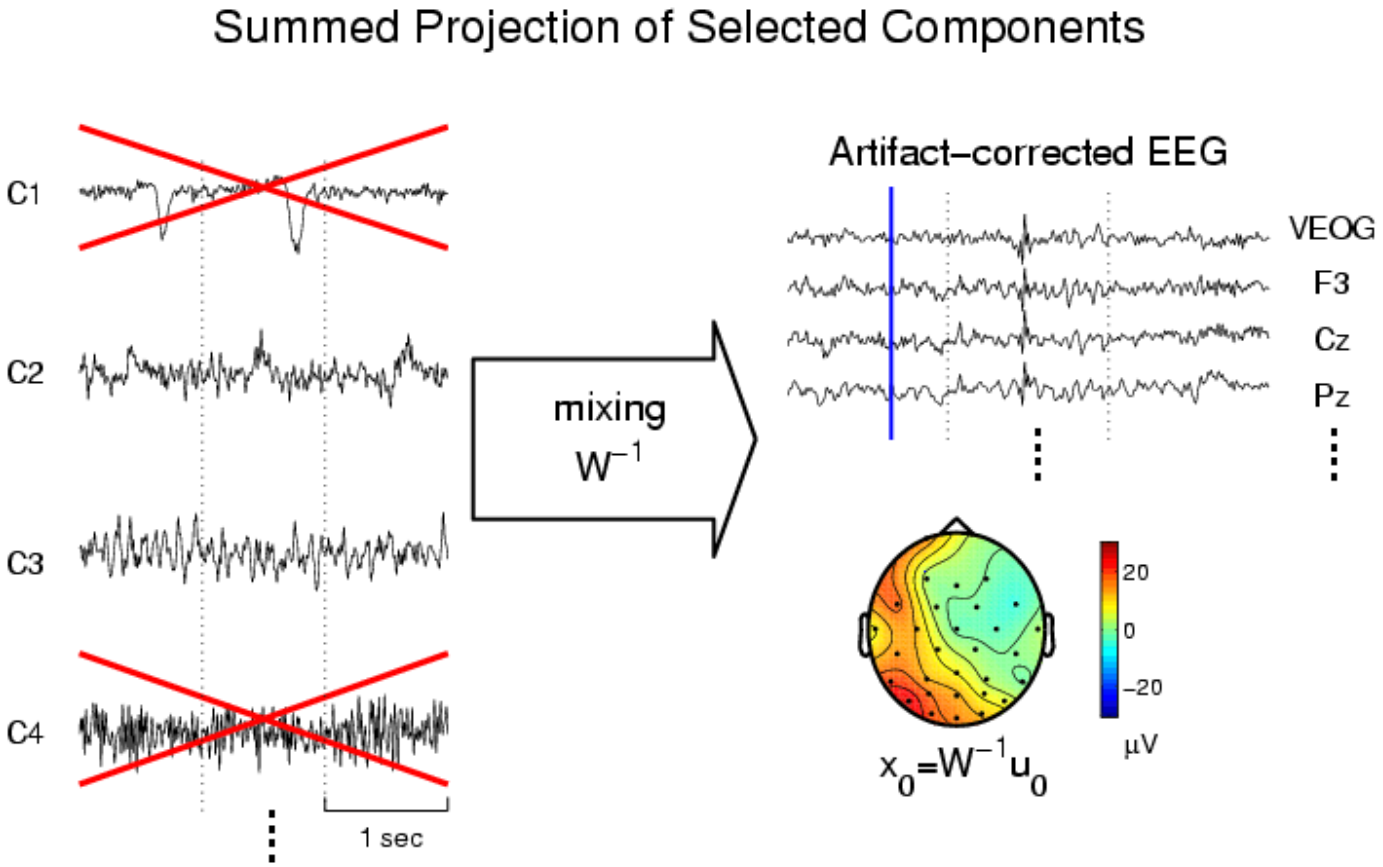
- Multiple sound sources in room (independent)
- Multiple sensors receiving signals which are mixtures of a few original sources
- Estimate original source signals from mixture of received signals
- Can be viewed as Blind-Source Separation as mixing parameters are not known

Examples in Healthcare: EEG Source Separation

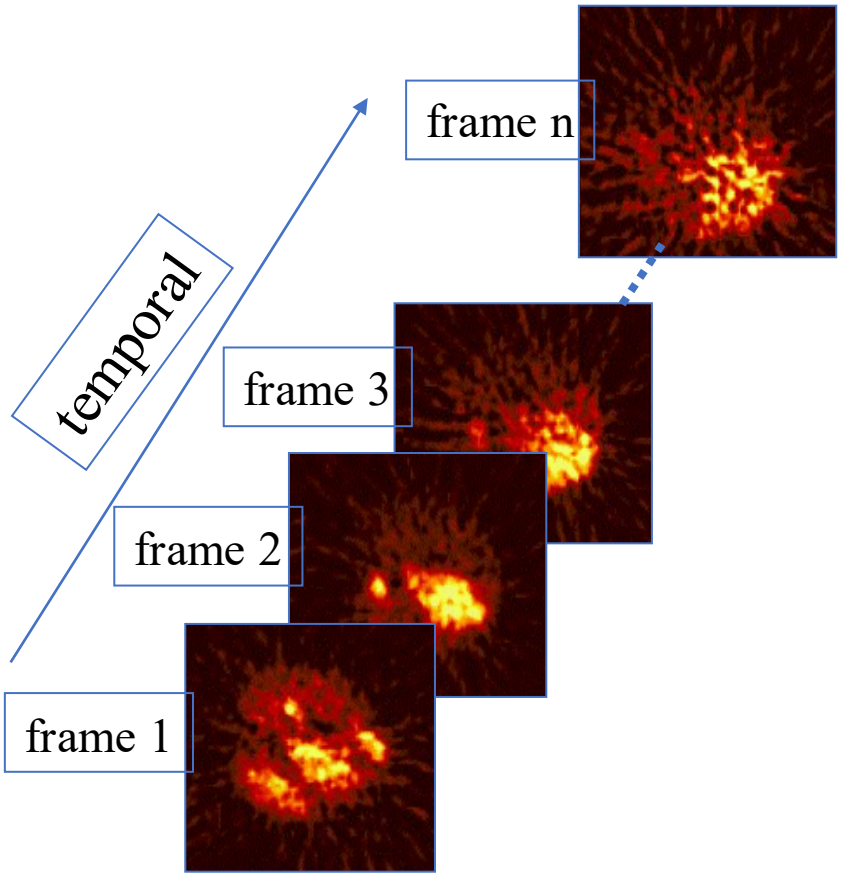




EEG Artifact Correction

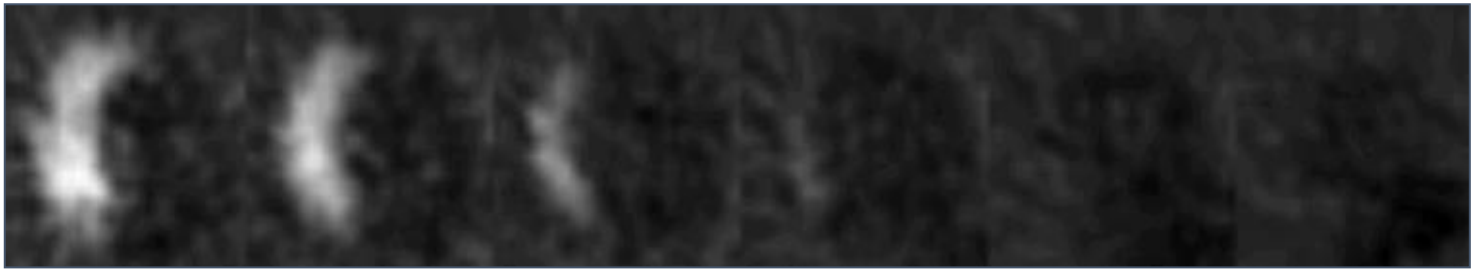


Dynamic Image Acquisition Using PET

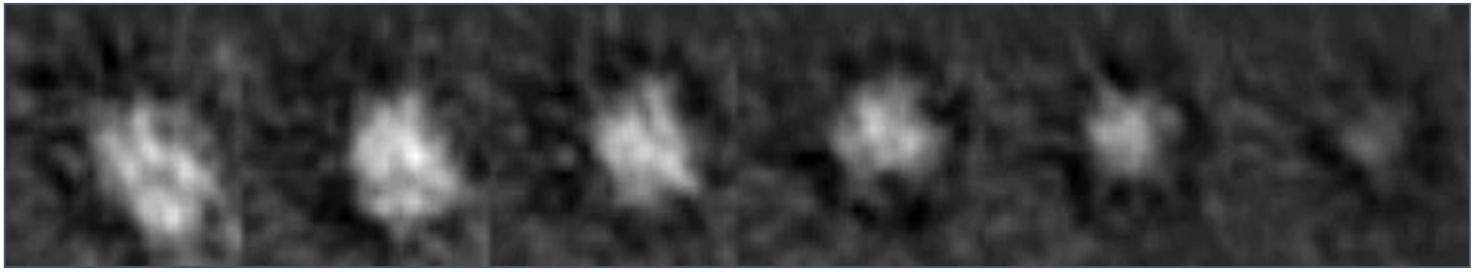


Independent Components

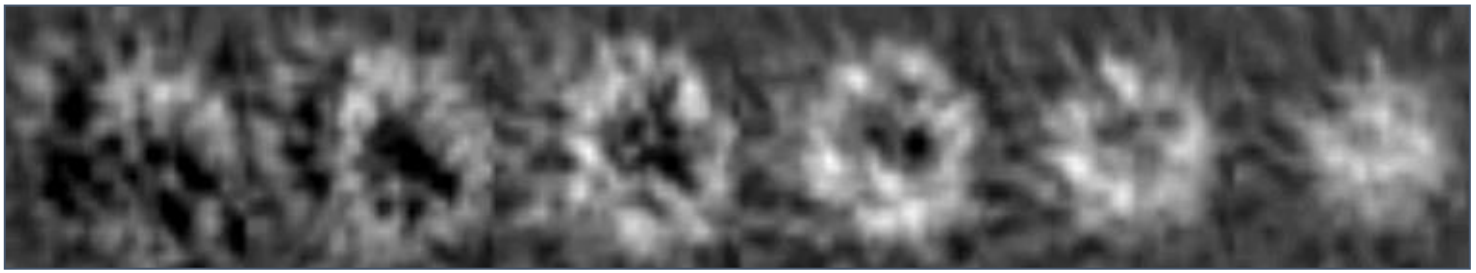
Right Ventricle



Left Ventricle



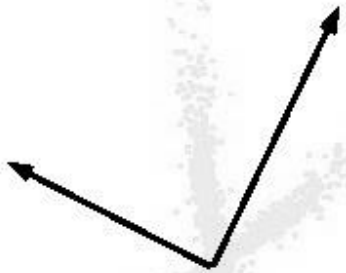
Tissue



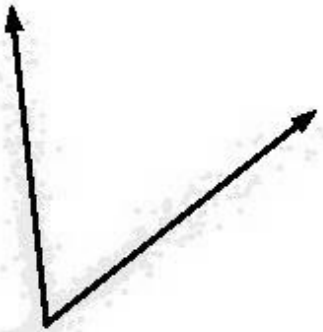
PCA vs ICA

- Linear Transform
- PCA
  - Focus on uncorrelated components (first order)
  - Some components are more important than others (defined by variance)
  - Orthogonal transformation
- ICA
  - Focus on independent components (higher order)
  - There is no notion of importance in the identified components
  - Non-orthogonal transformation

Independent Component Analysis (ICA)



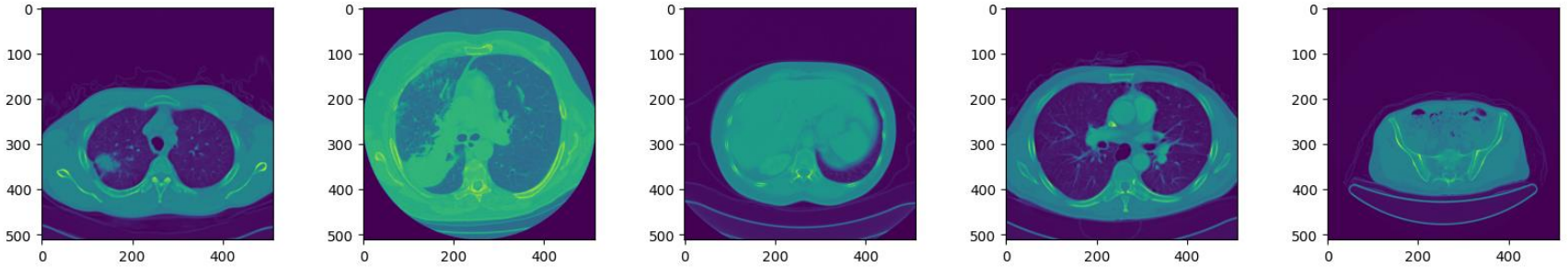
PCA
(orthogonal coordinate)



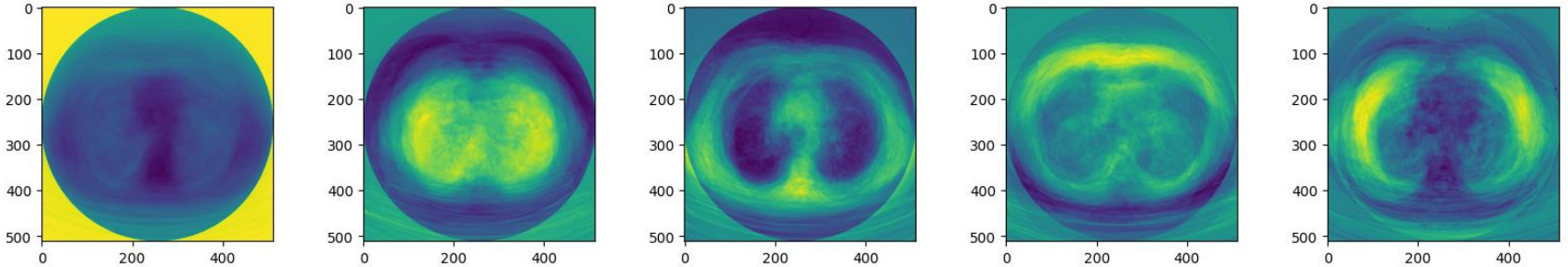
ICA
(non-orthogonal coordinate)

PCA vs ICA Examples

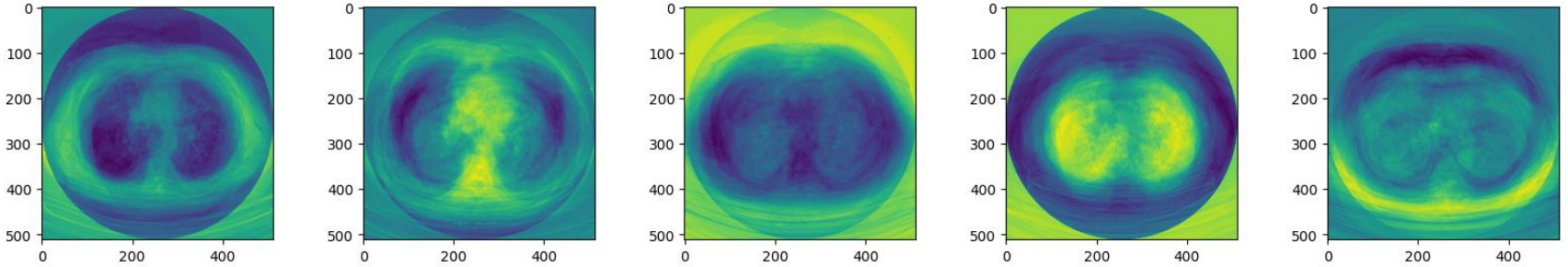
Original



PCA

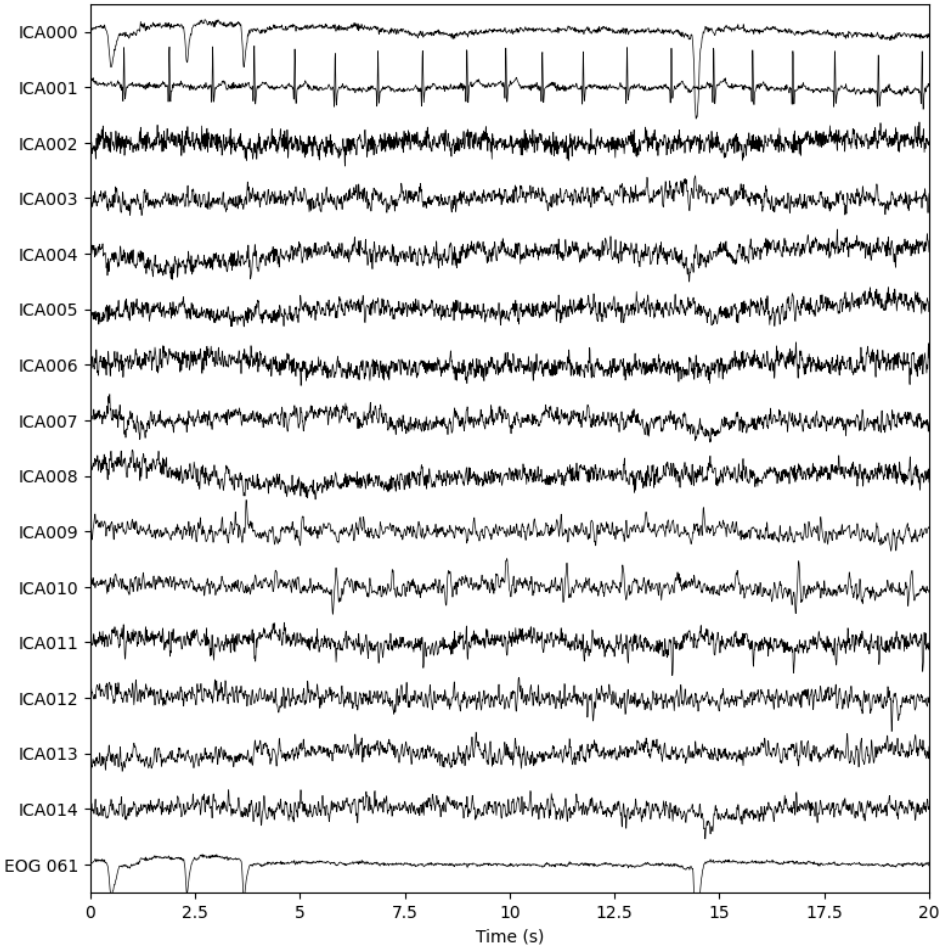


ICA

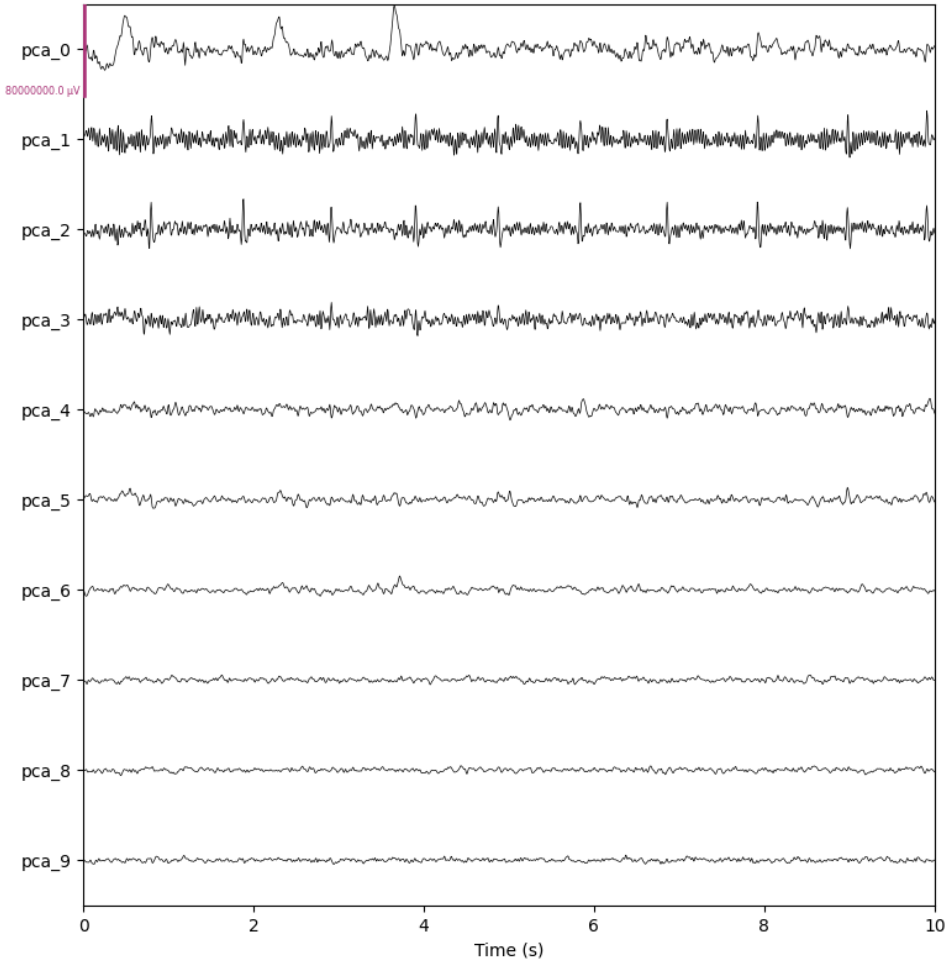


PCA vs ICA Example

ICA



PCA



ICA Applications

- Timeseries artifact removal
  - EEG, ECG
- Source separation
  - EEG, ECG, microarray data
- Medical image segmentation
  - CT, MRI

Exit pass

https://z.umn.edu/ml4h_exitpass

